Editor's comments:

The reviewers are in strong agreement that the current scope of the study is too narrow and the presentation too one-sided for publication in mSystems. Both recommend an expansion of the study to encompass both more datasets and more metrics of success, and suggest adopting a more neutral tone with a focus on when rarefaction is most appropriate and when it may not be.

If these changes are beyond the scope of the study the manuscript may be more appropriate for Spectrum but would still require some revision and review.

I appreciate the reviews provided by the two reviewers. Reading through my responses to the reviewers you will find that I was not swayed by the reviewers suggestions to make radical changes to my manuscript to broaden its scope. Because Martino was originally published in *mSystems*, I decided to submit my reproduction of their work to *mSystems*. The goal of my study was to reassess Martino's RPCA method using their original simulation data and expand their analysis to include rarefaction and other normalization and compositional data analysis methods. This was made clear in the Abstract and Introduction of the manuscript. Although the reviewers suggest a number of other analyses and a desire for me to use more simulation frameworks, I believe these are either well outside of the scope of what Martino did or duplicative of two recent papers I published (Schloss 2023, 2024). The reviewers sought a "comprehensive" study to settle the question of rarefaction once and for all. This is an unrealistic expectation that has never been achieve elsewhere in science.

I do not agree with the reviewers' claims that my analysis was one-sided. If anything I presented "other sides" by using the simulation framework of proponents of compositional methods, replicated their results, and then added a more comprehensive analytical framework by using additional methods to controlling for uneven sampling effort. The original Martino manuscript was so one-sided that I am not sure how they could say I was being one-sided given my attempts to strengthen the Martino study.

If you do not feel that my contributions are significant enough to be published in the journal, I will understand. I ask that a decision be rendered as soon as possible so that I can transfer the manuscript to either *mSphere* or *Microbiology Spectrum*.

The line numbers I reference correspond to the marked up version of the resubmitted manuscript.

Reviewer #1

This manuscript revisits the long-standing debate on rarefaction vs normalization/compositional approaches for uneven sequencing depth and concludes that rarefaction is superior. The author re-analyzes Martino et al.'s RPCA benchmarking, reports coding and methodological issues, modifies the simulations, and re-runs comparisons, ultimately concluding that rarefaction "remains the preferred method." The author has made valuable contributions in recent years advocating for rarefaction through systematic reanalysis of prior benchmarking studies (Schloss, 2023, 2024.), establishing rarefaction's practical utility. While the paper raises important reproducibility concerns and provides useful re-analysis, the current manuscript reads primarily as a targeted rebuttal rather than a balanced and comprehensive benchmarking study. Here are the major concerns

As stated at L129-135, the goal of my paper was to use the benchmarking framework originally used by Martino et al to compare their RPCA method to other normalization and compositional analysis methods as well as rarefaction. The reviewer is correct that this is a targeted rebuttal. I contend that the analysis is balanced. I have reused Martino's original simulation framework and then improved the framework by fixing the numerous coding errors present in their benchmarking code. This study was not intended to be comprehensive. Rather it adds to the two papers that the reviewer cited in this comment. A comprehensive benchmarking study is well beyond the scope of a single study. It is more thorough than Martino's analysis, which cherry-picked analysis approaches that are either not used in the microbiome field (e.g., KNN analysis) or that put RPCA in a more favorable light (e.g., using datasets with large effect sizes).

1. The manuscript addresses a focused methodological question through a reanalysis and modification of a previously published benchmark (Martino et al.). While this re-evaluation is useful and technically sound, the overall scope and impact may be limited. The study primarily revisits an existing simulation framework, identifies implementation issues, and evaluates rarefaction under revised settings using a relatively narrow set of tasks (e.g., PERMANOVA power, KNN classification, and sequencing-depth sensitivity). As currently framed, the work reads more as a methodological clarification or refinement of a prior benchmark than as a broadly impactful advance for the microbiome field. To strengthen the manuscript's impact, the evaluation should be expanded to include a broader range of common microbiome analyses—such as differential abundance testing, diversity estimation, distance metrics, regression models, multivariate inference, and count-based modeling—so that the conclusions are better supported across the wider microbiome workflow.

Because of the nature of RPCA, this really only makes sense in terms of beta diversity analysis. The "relatively narrow set of tasks" are the same tasks used in Martino et al. The other tasks the reviewer suggests have been done by others and myself in previous studies. The current study is already quite long to add other tools that do not necessarily relate to downstream applications of RPCA. Furthermore, the focus of this study was on the question of why I found one result using RPCA in another simulation and benchmarking framework and Martino found a different result.

2. The manuscript is currently framed more as a rebuttal to a specific prior study than as a neutral benchmark. The narrative repeatedly focuses on identifying issues in Martino et al.'s work (e.g., implementation details, simulation choices, and comparison setup), which shifts the focus toward a single-case critique and limits the breadth of evidence needed to support general conclusions. A robust benchmarking study would ideally include multiple methods, diverse simulation frameworks, and evaluations across a range of assumptions to assess generalizability. In contrast, the present work centers on critiquing one benchmark, replacing it with a revised simulation framework, and drawing broader conclusions. This

narrower design limits the generalizability of the findings and suggests that a more balanced, multi-method benchmarking approach would strengthen the manuscript.

I am not sure that there is such a thing as a “neutral benchmark”. However, I used the same simulation and benchmarking strategy used by Martino. I then improved it by fixing the numerous technical problems in their implementation so that it more closely reflected the approach that they described in their written text. In both cases, I found similar results: rarefaction outperformed RPCA and other compositional methods. These problems are not a matter of personal preference or style, they are fundamental problems such as inadvertently transposing matrices (or not), using nonsensical hyperparameters, and other issues that are described in the detailed code review provided in the Supplemental Text document. As outlined in my manuscript, this study is the third in a series that have all used different benchmarking approaches. All have come to a similar conclusion that rarefaction outperforms other methods when applied to alpha and beta diversity metrics.

3. The revised simulation framework still reflects a single modeling perspective. Although the manuscript critiques the simulations of Martino et al. and replaces them with a modified design, this does not remove simulation dependence but rather substitutes one set of assumptions for another. Key parameters, such as the number of features, the number of differential taxa, and calibration of effect sizes to achieve a specific PERMANOVA power, represent design choices that can strongly influence comparative performance. A comprehensive benchmarking study would ideally evaluate multiple simulation paradigms, including established microbiome simulation tools (e.g., MIDASim, GUniFrac-based simulations), along with a range of effect sizes, sparsity levels, and compositional structures. As currently presented, the results primarily reflect conclusions under a single simulation framework, which limits the generalizability of the findings.

My goal was not to remove simulation dependence in my re-analysis of the RPCA study. The purpose of the revisions to the simulation framework were to implement what the authors actually described in their manuscript (e.g., homo and heteroscedastic noise) and to resolve clear errors in

their code. Using yet another framework to evaluate RPCA and the other methods would not have answered the question of why Martino et al found one result using RPCA and I found a different one. By using Martino's simulations with the same broad set of methods I used in previous studies I was able to see that even with their original simulation data I found that RPCA did not perform as well as other methods. I was also able to identify coding errors in the code to simulate and analyze their data.

4. The manuscript demonstrates careful attention to detail in examining the statement that "the most notable difference to Martino's version of the figure was that they inadvertently switched the labels of the axes and performed the One CLR transformation on features rather than samples." However, this interpretation may be somewhat over-scrutinized. Differences in data orientation are very common between statistics and biology: statistical workflows typically place samples in rows and variables in columns, whereas biological pipelines often arrange features in rows and samples in columns. Depending on the input format used in Martino's implementation, their approach may simply reflect the statistical convention and therefore may not require transposing the data. This point should ideally be confirmed with the authors of the original paper before drawing firm conclusions.

This is not a matter of difference in data orientation. As stated in my manuscript, the authors inadvertently switched the labels of the axes and performed the One CLR transformation on features rather than samples. This is elaborated upon in the Supplemental Text code review where I point to the specific code on page 2 of the supplement. As presented it appears that features overlap across samples. In reality, samples overlapped across features - this doesn't make sense. Unfortunately, this error propagated through the rest of their simulation data. By working through the code and data outputted at each step, as I did in the Supplemental Text, it is clear this is not a matter of statistical convention.

5. This statement in line 155 appears overly critical. In practice, focusing on the first two axes is a common and reasonable convention in microbiome analyses because ordination plots are primarily used for visualization. When group differences are strong, they typically

appear in the leading axes for PCA/PCoA-type methods. If signals emerge only in higher dimensions, this more likely indicates a weaker or more diffuse effect rather than a methodological flaw. RPCA does not inherently require the use of two axes; the rank is a user-defined parameter similar to the number of principal components in PCA. From this perspective, the original choice to focus on two axes does not constitute an error. Instead, the present manuscript's extension to additional axes can be viewed as a useful expansion that explores additional scenarios. However, this extension is limited in scope and largely represents a comparison with a single method, which reduces the broader impact of this point.

It is unclear to me what in L155 is overly critical. My sentence states that my result agreed with their result, "Replicating what Martino reported, the first two axes of the Bray-Curtis ordinations only explained a fraction of the total variation in the data while the RPCA ordinations explained all the variation." As I go on to describe later in the manuscript, increasing the number of axes for RPCA results in significant changes to the analysis (e.g., Figure 7). I highlight that there is no objective justification for 2 or 3 or 9 axes which may lead to one selecting a number of axes that gives rise to a desired result (L397-402).

6. Some researchers view compositional transformations as an alternative to rarefaction rather than a preprocessing step that must be applied after rarefaction. In the compositional data analysis framework, library size differences are treated as a scaling factor, and methods such as CLR or other normalization approaches are intended to account for this directly. From this perspective, applying rarefaction prior to CLR may be considered redundant, as both steps are intended to address sequencing depth effects, albeit from different methodological viewpoints. This broader context may also help explain why Martino et al. did not apply rarefaction prior to RPCA. Their work is grounded in the compositional data analysis paradigm, which assumes that sequencing counts are compositional and that differences in library size can be addressed through normalization or log-ratio transformations rather than subsampling. From this perspective, the absence of

rarefaction is a methodological choice consistent with that framework rather than an oversight.

I agree that combining rarefaction and compositional transformations is redundant. I am unable to find where in the manuscript I criticized Martino for not using rarefaction prior to RPCA. After all, The claim made by Martino et al. is that those transformations replace rarefaction and are not sensitive to differences in read depth. My criticism of Martino was that for Bray-Curtis calculations they rarefied their data to a 10-fold smaller depth than was necessary (L311-316). I went on to point out that the difference didn't really matter (L317-318).

7. In line 345, the statement that RPCA-based methods performed worse when the effect size was small should be interpreted cautiously. Reduced performance in low–effect size settings may reflect the well-known trade-off associated with log-ratio transformations, where variance stabilization and compositional correction can come at the cost of statistical power, particularly in sparse microbiome data. Reduced performance in low–effect size settings is not unexpected and may reflect the known trade-off associated with log-ratio transformations, which tend to shrink variability and attenuate contrasts, particularly in sparse microbiome data. As such, this observation does not necessarily indicate a flaw specific to RPCA, but may instead reflect broader properties of CLR-based approaches.

Although these observations from the reviewer make sense to me, I am unable to find a citation to them in the literature. If anything McMurdie & Holmes argue that rarefaction causes a more significant hit to power than the other methods they and I have used. If it is generally true that rarefaction has more power than the other methods (as I and the reviewer seem to agree), then I don't see what the argument for their use is since one does not know the effect size study *a priori*. If one suspects they have a small effect size, then why use a low-powered test? Because the analysis in this study is more focused on RPCA than CLR-based methods in general, I will limit their critique in this sentence to RPCA. I have added text at L393-395 that summarizes the need for researchers to select the most powerful tests available.

8. The rarefaction versus normalization debate has persisted for over a decade, originating from fundamentally different statistical philosophies. McMurdie & Holmes (2014) criticized rarefaction as "inadmissible," yet acknowledged that transformations can reduce power depending on model assumptions—a critical nuance often ignored. Weiss et al. (2017, Microbiome) demonstrated that normalization performance depends strongly on data characteristics, with no single method universally superior. Despite this established complexity, the current manuscript fails to comprehensively address why this debate persists or how to resolve it. The analysis does not answer fundamental questions: Under what specific conditions (sparsity, effect size, library size variation) does each approach maintain valid error control and optimal power? How do rarefaction's theoretical limitations (no variance model, arbitrary thresholds, information loss) compare quantitatively to log-ratio transformations' known drawbacks in sparse data (Gloor et al., 2017, Front. Microbiol.)? By selectively benchmarking RPCA against rarefaction using metrics and datasets that favor depth-equalizing methods, rather than systematically characterizing when different approaches succeed or fail, the manuscript perpetuates advocacy rather than providing principled guidance. Both approaches have documented limitations. Rarefaction has been criticized for inducing bias and discarding data (Willis, 2019; McMurdie & Holmes, 2014), while normalization methods show data-dependent performance and can inflate false discovery rates (Weiss et al., 2017; Lin & Peddada, 2020). A comprehensive evaluation would focus on developing decision frameworks for method selection based on dataset characteristics. The long-standing debate will continue until the field moves beyond polarized positions and establishes clear, evidence-based guidelines for when rarefaction's empirical robustness versus normalization's statistical rigor is most appropriate.

The reviewer is correct that I don't state "under what specific conditions (sparsity, effect size, library size variation) does each approach maintain valid error control and optimal power". That's beyond the scope of this paper and I suspect beyond the scope of any single study. My goal was to use the simulations implemented by Martino et al. within a more comprehensive framework than they used

to see why they were so positive about their method and my previous analyses were less so. Along the way, I found significant problems in their benchmarking code. My analysis is far more comprehensive than the original Martino. Taken with my two other papers that the reviewer had cited in an earlier point, the analysis is now more comprehensive because it uses the simulation framework proposed by Martino. It joins the conversation with the other literature. I find the Reviewer's comments to be unfair when they appear to take the papers they cite as truth, when several of them are reviews (e.g., Lin & Peddada, Gloor) or do not actually provide any benchmarking (e.g., Willis). Meanwhile my previous papers appear to be discounted in this discussion. Science moves forward one step at a time correcting and adding to what has come before it. As an illustration of this the version of the manuscript that I submitted cited all of the studies that the reviewer cited. A single comprehensive study that satisfies everyone has never been proposed in any area of science and is unreasonable to expect here.

Reviewer #2

This manuscript challenges the prevailing compositional-data paradigm by arguing that rarefaction can outperform RPCA and related methods in controlling uneven sequencing depth. The author presents extensive benchmarking across the Martino et al. datasets and additional datasets, which is valuable. However, several conceptual, methodological, and presentation issues should be addressed before the conclusions can be fully supported.

1. Although rarefaction outperforms compositional methods in certain scenarios, the manuscript draws a very strong conclusion that "Rarefaction remains the preferred method." This framing is unnecessarily polarizing and overlooks the contextual nature of normalization choices. The more relevant question is not which method is universally superior, but under which conditions each method is appropriate. The author should explicitly discuss scenarios such as: a) part of samples with low sequencing depth or sparse counts; b) strong batch effects and cross-study heterogeneity in amplicon datasets; c) variability in rRNA gene copy number across microbes.

A nuanced decision framework (rather than a universal recommendation) would significantly improve the manuscript's impact and fairness.

The reviewer raises a number of interesting conditions that could be included. Although I believe that their suggestion (a) is covered in this and my previous analyses, I am reluctant and unsure of how to account for (b) and (c). I am not familiar with arguments for using compositional methods when there are batch effects or variation in rRNA copy number. There would appear to be other approaches to correct the underlying counts before downstream processing by rarefaction or any of the other methods that I used. Because these conditions were not included by Martino and my goal was to reproduce and strengthen their analysis, I believe these considerations are outside of the scope of my study. Frankly, in my three studies, I have yet to identify a nuanced decision framework - rarefaction is the most reliable method for controlling for uneven sampling effort. Other methods are at best as good as rarefaction and often much worse. The safest choice would be to use rarefaction regardless of the dataset.

2. The manuscript primarily evaluates preprocessing methods using classification and PERMANOVA. However, microbiome studies also aim to identify biomarkers, infer ecological structure, or construct co-occurrence networks. Rarefaction and CLR-based transformations can affect these downstream analyses differently. At minimum, the author should discuss potential implications for biomarker discovery and network inference, or ideally provide benchmarking results for at least one such downstream task.

This study was designed to reproduce Martino and build upon it by adding additional methods of correcting for multiple comparisons within an an analysis framework based on beta diversity metrics. Other studies of mine (Schloss 2023 and 2024) have looked at alpha and beta diversity as well. Other studies that I cited look at biomarker discovery (e.g., Weiss et al). Considering the study is already quite long and biomarker discovery or network inference were not included in Martino, I am reluctant to add them to my reproduction of their analysis.

3. In datasets with unequal group sizes (e.g., the sponge dataset), accuracy alone is not an

adequate performance metric. AUROC, AUPRC, or balanced accuracy would provide a more comprehensive assessment. The manuscript should report at least AUROC for imbalanced scenarios.

In the analysis performed by Martino they used a balanced design by randomly subsetting the number of samples from each treatment group. Furthermore, the original Martino study used accuracy as the metric to assess classification performance. To keep the analysis parallel between my study and theirs, I think it is warranted to use accuracy. I don't think much is gained by using an alternative metric.

4. The author states that weighted UniFrac was not included due to difficulty in generating reliable trees and because Bray-Curtis and weighted UniFrac rarely disagree (but no reference was cited in the manuscript). Since all sequences were mapped to the SILVA database, which provides a reference phylogeny. Therefore, UniFrac distances could be computed using the SILVA tree.

I have added the weighted UniFrac analysis requested by the reviewer (introduced at L299-303). Through integrating this into my analysis, I found an additional problem with Martino's analysis of their case studies. Namely, the trees that they supplied did not include all of the sequences in their OTU table. As a result, they removed the missing sequences from the OTU table. Unfortunately, the filtered OTU table was not used to run RPCA. The result was another apples-to-oranges comparison between the Bray-Curtis and weighted UniFrac analyses and the RPCA analysis. Text describing this has been added to the Manuscript (L283-285) and the Supplemental Text (P16, paragraph beginning with "Third, in my attempts to reproduce the weighted UniFrac distances. . . ").

5. The analysis focuses on amplicon OTU/ASV datasets. It remains unclear whether the conclusions extend to shotgun metagenomic profiles, which have different count distributions and compositional properties. The author should either provide additional analyses or explicitly discuss the expected applicability and limitations in shotgun metagenomics.

This manuscript only discusses amplicon sequence data. The only mention of metagenomic data is in the opening sentence of the paper to state that uneven sampling effort is a significant problem for both approaches. I have opined on the need for rarefaction of metagenomics data in other studies, but do not feel it is necessary to do so here since this study is an attempt to reproduce and build upon the simulations and analysis of Martino et al.

6. For multi-panel figures (e.g., Figures 1 and 3), each panel should be explicitly described in the figure legend. This will improve readability and reduce ambiguity for readers and reviewers.

Thank you for catching this oversight. The lettered panels have now been described explicitly in the figure caption.

7. This manuscript adopts a strongly adversarial tone toward Martino et al., using phrases such as "straw man," "critical errors," and "foreign to standard microbiome analyses." While scientific critique is appropriate, such language may be perceived as polemical and could detract from the manuscript's reception. I recommend moderating the rhetoric and focusing on technical arguments rather than adversarial framing.

I suspect we all have different tolerances for tone of language. For instance, "inadmissable" was used in the title of the McMurdie & Holmes study that started the debate over rarefaction. I consider that word to be needlessly inflammatory and incorrect. Regardless, I have removed "straw man" from the text. I believe that "foreign to standard microbiome analyses" was the most charitable way of describing analyses that have yet to be used in microbiome studies (e.g., KNN classification). I believe this is a more tolerable statement than simply saying "wrong." Similarly, the errors that I outlined in the extensive detailed code review caused critical problems in the simulations that caused them to deviate significantly from what the authors described in the text of their papers. These errors were critical.

8. In addition, I also suggest to change the article type from "Research Article" to "Methods and Protocols" or "Opinions/Hypotheses".

I do not see how the manuscript I submitted would fall under either of these categories. It is not a method or protocol and is not an opinion or hypothesis. If *mSystems* has a format like *Nature*'s "Matters Arising", I would consider that but suspect it is too long for such a format.